

A QUBO Formulation for Flight-Trip Sequencing with Start-Break Variables and k -Local-to-Quadratic Reduction

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Abstract

We present a Quadratic Unconstrained Binary Optimization (QUBO) formulation for a flight-trip sequencing problem that arises in airline crew scheduling. The model assigns flight segments to ordered positions while permitting controlled trip breaks through explicit start-flag variables. Operational feasibility and cost are encoded using a directed transition graph capturing station continuity, connection-time rules, base constraints, and check-in/check-out effects. The start-break mechanism naturally induces higher-order (k -local) interactions when a start suppresses a previously-applied pairwise gap cost. We describe an explicit quadratization strategy based on ancilla substitutions with penalty enforcement, yielding a purely quadratic model compatible with classical samplers, hybrid solvers, and quantum annealing backends. The formulation is designed to be transparent, modular, and practical for experimentation on realistic schedules, offering a bridge between airline trip-construction logic and emerging quantum optimization workflows. This paper documents the mathematical formulation only; it is a companion to a separate Python reference implementation and is not intended as an empirical benchmarking study.

Keywords: QUBO, BQM, crew scheduling, trip construction, sequencing, quadratization, quantum annealing, hybrid optimization

1 Introduction

Trip construction (pairing generation) is a central component of airline crew scheduling. In practice, trip feasibility depends on station continuity, minimum and maximum connection times, base start and return rules, and operational costs such as sit time and duty setup/teardown. Classical approaches frequently leverage heuristics, dynamic programming, or MILP-based methods embedded in decomposition frameworks.

QUBO models provide a unified representation consumable by a range of solvers: simulated annealing, tabu search, hybrid solvers, and quantum/hybrid backends. The main challenge is encoding realistic trip rules and conditional cost logic into a quadratic form. This paper presents a QUBO formulation that (i) maintains an explicit ordered assignment structure, (ii) introduces start-break variables to represent trip boundaries, (iii) derives feasibility/cost from a graph representation of permissible transitions, and (iv) reduces k -local terms introduced by start-mediated cancellation into a quadratic model via ancillas.

Motivation and Intent The QUBO formulation here was developed specifically to make trip construction solvable on a D-Wave quantum annealer, which requires problems to be expressed in this quadratic, binary form.

It is not proposed as a better way to solve static trip construction on classical computers: mature classical trip generators already perform well for this problem, and reformulating it as a QUBO offers no particular advantage when run classically. The interest of this formulation is specifically as a bridge to quantum annealing hardware; classical solvers appear in this paper only as reference baselines for validating the QUBO model itself, not as a suggested use case.

The outlook is different for Day-of-Operations recovery — re-sequencing crew and trips within minutes under active disruption — a considerably harder, more time-critical regime than the static problem addressed here, and the one where this approach’s case is strongest.

Scope This paper documents a mathematical formulation only. It is written as a companion reference to an existing Python implementation and does not include computational results, solver benchmarks, or qubit/embedding counts for specific problem sizes; those are treated separately. Likewise, crew legality rules beyond basic sequencing and connection feasibility (duty time, minimum rest, time-zone handling) are out of scope here and are discussed as future extensions in Section 11.

2 Problem Definition

Given a set of flight segments

$$F = \{1, \dots, N\},$$

we seek an ordered assignment of these segments into sequence positions

$$R = \{1, \dots, N\},$$

while allowing the sequence to be partitioned into multiple trips by inserting starts.

Goal. Minimize total operational cost subject to:

- Each position holds exactly one segment (row uniqueness).
- Each segment appears exactly once (column uniqueness).
- Consecutive segments within a trip satisfy feasibility (station continuity and connection-time rules).
- Trip starts and ends respect base constraints, with check-in/check-out effects.

This formulation focuses on sequencing and trip boundary placement. It is intended to serve as a reusable kernel that can be embedded in broader pairing/rostering workflows.

3 Variables and Core Representation

3.1 Assignment Grid

Binary variables

$$x_{r,n} \in \{0, 1\}, \quad r \in R, \ n \in F$$

indicate that flight segment n is assigned to sequence position r .

3.2 Start-Break Column

Binary variables

$$s_r \in \{0, 1\}, \quad r \in R$$

indicate that position r begins a new trip. Informally, $s_r = 1$ breaks continuity between positions $(r - 1)$ and r . Position $r = 1$ has no predecessor, so the model strongly biases s_1 toward 1 (the first occupied position begins a trip) using an asymmetric penalty term rather than fixing it as a hard-coded constant; s_1 therefore remains, in principle, a free binary variable that is energetically favored to take the value 1 rather than one pinned by construction. This bias, combined with the absence of a predecessor at $r = 1$, is what removes the undefined-boundary case addressed in Section 5.2.

3.3 Ancillas for Quadratization

Additional binary variables $y_k \in \{0, 1\}$ are introduced during reduction of higher-order terms to quadratic form (Section 8).

3.4 Visual Summary

Figure 1 shows the variable layout: an $N \times N$ assignment grid plus a side column of start variables.

	f_1	f_2	$\cdots$	f_{N-1}	f_N	Start
r_1	$x_{1,1}$	$x_{1,2}$	$\cdots$		$x_{1,N}$	s_1
r_2						s_2
r_3			$x_{r,n}$			s_r
r_4						
r_5						s_N

Table 1: Conceptual QUBO variable layout: assignment grid $x_{r,n}$ with a dedicated start-break column s_r . One “1” per row and per column enforces a permutation-like assignment; $s_r = 1$ indicates position r begins a new trip (break between $r - 1$ and r).

4 Structural Constraints (Permutation-Like Assignment)

We enforce the assignment structure using quadratic penalties.

4.1 One Segment per Row

For each position r :

$$\sum_{n=1}^N x_{r,n} = 1.$$

QUBO penalty:

$$H_{\text{row}} = \lambda_{\text{row}} \sum_{r=1}^N \left(\sum_{n=1}^N x_{r,n} - 1 \right)^2.$$

4.2 One Row per Segment

For each segment n :

$$\sum_{r=1}^N x_{r,n} = 1.$$

QUBO penalty:

$$H_{\text{col}} = \lambda_{\text{col}} \sum_{n=1}^N \left(\sum_{r=1}^N x_{r,n} - 1 \right)^2.$$

These constraints produce only linear and quadratic terms after expansion, making them QUBO-native.

5 Graph-Based Transition Feasibility and Cost

We model feasibility and operational costs via a directed graph G over segments (and conceptual start/end behavior).

5.1 Segment-to-Segment Edges

For segments $i \rightarrow j$, an edge exists if:

- station continuity holds (arrival of i equals departure of j),
- the connection time gap satisfies minimum and maximum thresholds,
- any additional base/operational rules are satisfied.

If allowed, the edge receives a weight g_{ij} capturing sit time and endpoint adjustments.

5.2 Start/End Effects (Base Check-in/out)

Start-related costs are represented by terms that connect s_r to the segment selected at row r (check-in) and to the segment at row $(r - 1)$ (check-out). Because position $r = 1$ has no predecessor (Section 3.2), the check-out term is defined only for $r \geq 2$:

$$H_{\text{CI}} = \sum_{r=1}^N \sum_n c_n^{\text{in}} s_r x_{r,n}, \quad H_{\text{CO}} = \sum_{r=2}^N \sum_n c_n^{\text{out}} s_r x_{r-1,n}.$$

These are quadratic since they involve products of two binary variables. Restricting the check-out sum to $r \geq 2$ removes the otherwise undefined reference to $x_{0,n}$.

Note on weight granularity. For clarity, c_n^{in} and c_n^{out} are presented above as single per-segment weights. In practice each weight is computed from a more detailed rule that distinguishes whether

the segment’s relevant endpoint is at a home base or an away station, and whether the position represents a departure or a return leg – i.e. up to four distinct weight components feed into a given c_n^{in} or c_n^{out} value depending on context. The formulation above remains correct at the level of the QUBO structure; the added detail only affects how the scalar weight itself is computed, not the quadratic form it enters.

Note on the first occupied position. At $r = 1$ there is no preceding position, so there is no check-out term to cancel there (Section 5.2). Instead, the implementation cancels a separate placeholder cost that is otherwise charged for leaving a segment unsequenced, replacing it with the check-in term once row 1 is assigned. This is a boundary-handling detail of the same kind as the $r \geq 2$ restriction above, not an additional modeling rule.

5.3 Misconnect Penalties

If a consecutive-row pairing $(r, r + 1)$ selects a segment pair (i, j) for which no feasible edge exists, a penalty is applied:

$$H_{\text{mis}} = \lambda_{\text{mis}} \sum_{r=1}^{N-1} \sum_{i,j} M_{ij} x_{r,i} x_{r+1,j},$$

where $M_{ij} = 1$ indicates an invalid transition.

6 Objective Terms (Gaps and Preferences)

A typical objective adds gap costs for valid consecutive transitions, start-related costs (check-in/out) at trip boundaries, and bias terms to encourage reasonable starts or discourage undesired patterns. A simplified pairwise gap objective is:

$$H_{\text{gap}} = \sum_{r=1}^{N-1} \sum_{i,j} g_{ij} x_{r,i} x_{r+1,j}.$$

7 k -Local Interactions Induced by Start Breaks

When a start variable suppresses a pairwise gap cost between rows $(r - 1)$ and r , the natural algebra becomes cubic:

$$\Delta H \propto -g_{ij} x_{r-1,i} x_{r,j} s_r.$$

These k -local terms arise from conditional logic: apply the gap cost only if the boundary is not a start.

	f_1	f_2	f_3	f_4	f_5	s_r
r_1			•			1
r_2					•	0
r_3	•					0
r_4				•		1
r_5		•				0

Table 2: Concrete example: selected assignments (dots) and start flags. Starts at r_1 and r_4 create a trip break between r_3 and r_4 , motivating conditional (k -local) terms before quadratization.

8 Quadratzation: k -Local to Quadratic via Ancillas

To solve on QUBO/BQM backends, higher-order terms must be converted to quadratic form. The substitution used here follows the standard pseudo-Boolean reduction technique introduced by Rosenberg [2] and formalized by Boros and Hammer [3], and is consistent with the reduction utilities documented for D-Wave’s Ocean SDK [4].

8.1 Cubic Reduction Rule

For a cubic term $Wx_1x_2x_3$, introduce an ancilla y intended to represent $y = x_1x_2$. Enforce consistency using a penalty:

$$H_{\text{cons}} = P(x_1x_2 - 2x_1y - 2x_2y + 3y).$$

Replace the cubic term with:

$$Wx_1x_2x_3 \mapsto H_{\text{cons}} + Wyx_3.$$

For binary variables, the penalty is minimized when $y = x_1x_2$; choosing P sufficiently large (in practice, P greater than the largest-magnitude coefficient W appearing in any reduced term) preserves the original objective ordering among feasible assignments.

8.2 Higher-Order Terms

For a k -local product $W \prod_{\ell=1}^k z_\ell$, we iteratively:

1. select a pair (z_1, z_2) and introduce (or reuse) $y = z_1z_2$,
2. apply a consistency penalty for newly introduced y ,
3. replace the product with $Wy \prod_{\ell=3}^k z_\ell$,
4. repeat until only quadratic terms remain.

Reusing substitutions reduces ancilla growth and improves hardware portability.

9 Weight Hierarchy and Scaling

The full energy is a sum of components:

$$H = H_{\text{row}} + H_{\text{col}} + H_{\text{mis}} + H_{\text{gap}} + H_{\text{CI}} + H_{\text{CO}} + H_{\text{cons}} + \dots$$

Weights are chosen in tiers so that structural constraints dominate objective improvements, ancilla consistency penalties dominate any incentive to violate substitutions, and feasibility penalties dominate gap-reduction incentives that would otherwise create misconnects.

In practice, a simple and effective starting heuristic is to set each constraint weight to exceed the largest possible objective-term contribution it could be traded against — for example, $\lambda_{\text{row}}, \lambda_{\text{col}}, \lambda_{\text{mis}} \gg \max_{i,j} |g_{ij}|$, and $P \gg \max |W|$ for every reduced ancilla term — then tightened empirically by

monitoring constraint violations on sampled solutions. This is a tuning procedure rather than a closed-form rule; exact values are instance-dependent and are addressed in the accompanying implementation rather than in this formulation paper.

10 Solvers and Portability

Once quadratized, the model is a standard QUBO/BQM and can be solved using simulated annealing and tabu search baselines, hybrid solvers, and quantum annealing backends (subject to embedding and hardware limits). This portability is useful for benchmarking and for exploring which problem sizes and structures benefit from different solver regimes.

11 Discussion

The formulation emphasizes clarity:

- Operational rules are represented explicitly via a transition graph.
- Variable semantics are visually clear (Figures 1–2).
- Conditional logic is handled through start-break variables, with a direct explanation for k -local terms.
- Quadratzation is explicit, reusable, and implementable with ancilla reuse, following the standard Rosenberg/Boros–Hammer substitution.

Incorporating additional crew legality rules (duty time, minimum rest, time-zone handling) is intended future work. Whether such rules can be added by refining graph construction and objective terms while preserving the same QUBO structure, or whether they require a different modeling approach, has not been established and will require further investigation and validation. These extensions, along with computational results and problem-scale (qubit/variable count) analysis, are left to future work and to the accompanying implementation rather than this formulation paper.

12 Conclusion

We presented a QUBO formulation for flight-trip sequencing that combines permutation-like assignment, explicit trip starts, graph-based feasibility/cost modeling, and a practical quadratzation pipeline for higher-order terms induced by conditional cost cancellation. The resulting quadratic model is compatible with a wide range of classical, hybrid, and quantum optimization backends.

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